

Manual testing

Day-07 Assignment

1.Explain the advantages and disadvantages of White Box Testing

Advantages and Disadvantages of White Box Testing

Advantages

- **Early Defect Detection:** Finds bugs during initial development phases like unit testing, lowering fixing costs.
- **Thorough Code Coverage:** Validates execution paths, statements, and conditions that external tests miss.
- **Code Optimization:** Exposes hidden bottlenecks, dead code, and memory leaks to streamline the codebase.
- **Security Enhancement:** Uncovers vulnerabilities, weak validation rules, and hardcoded credentials within the code.

- Easy Automation: Integrates cleanly into CI/CD pipelines since test scripts tie directly to functions.

Disadvantages

- High Complexity: Demands specialized programmers who deeply understand application architecture.
- High Maintenance Overhead: Rewriting code invalidates existing scripts, forcing testers to continuously update tests.
- Missed Requirements: Fails to capture missing business requirements if the logic was omitted entirely.
- No User Perspective: Focuses strictly on code internals, frequently missing usability or user interface (UI) issues.
- Scalability Bottlenecks: Testing every single path becomes exceptionally difficult and expensive as code scale grows

2.Explain about Path Coverage and Multiple Condition Coverage:

Path Coverage:

Every possible execution path from the start of a program to its completion is executed at least once. It provides a fine-grained, comprehensive analysis of a program's control flow.

Example

Consider this code block:

text

1. If ($A > 0$) Then
2. Print "A is Positive"
3. If ($B > 0$) Then
4. Print "B is Positive"

Multiple Condition Coverage:

Multiple Condition Coverage focuses exclusively on decision statements containing multiple sub-conditions combined by logical operators like AND or OR.

Example

Consider a single decision statement with three sub-conditions ($n = 3$):

text

If ($X > 0$ AND $Y > 0$ AND $Z > 0$)